INFRASTRUCTURE COST ANALYSIS: TRADITIONAL VS HYBRID

UNOFFICIAL DOCUMENT

EXECUTIVE SUMMARY

This cost analysis evaluates IT infrastructure options for a new bank branch serving approximately 20 employees, comparing traditional on-premises deployment ($149,964 CAD CapEx hardware + $66k-$78k labor) against a hybrid cloud model ($100,896 CAD CapEx + $54k-$63k labor). Hybrid delivers 33% lower Year 1 costs and projects $546k savings by Year 10 through reduced hardware, elastic scaling, and lower staffing (1.75 FTE vs. 2.0 FTE annually).

Key Financial Findings

- Traditional Year 1 total: ~$216k-$228k CAD; 5/10-year TCO higher due to refresh cycles and OpEx ($238k-$298k/yr).

- Hybrid Year 1 total: ~$155k-$164k CAD; superior long-term via OpEx shift to cloud ($215k-$269k/yr) and 20% less labor.

- Annual FTE labor: Traditional 2.0 ($238k-$298k); Hybrid 1.75 ($215k-$269k), assuming shared central IT.

Strategic Benefits

Hybrid enables faster rollout (4.5 months), AI-ready scalability, and compliance for sensitive workloads while keeping core banking on-premises. Traditional offers control but limits agility and innovation.

RECOMMENDATION

Traditional:

- Full control with data
- Straightforward regulatory compliance
- operational workflows are critical

Hybrid:

- Lower total cost of ownership
- Scalability
- Advanced Capabilities

INTRODUCTION AND SCOPE

This cost analysis report evaluates the infrastructure costs associated with opening a branch within the bank, supporting an internal population of approximately 20 employees.

The scope covers the setup of core IT services such as branch banking systems, file sharing, email, and collaboration tools while comparing a traditional on-premise deployment with a hybrid cloud approach.

The objective is to guide investment decisions for future branch expansions by highlighting differences in upfront capital, ongoing operational costs, and total cost of ownership over a 5-year to 10-year period.

Key Assumptions

- Costs in CAD (March 2026 vendor quotes)
- 20-employee steady state
- Shared central IT staffing (2.0 FTE traditional, 1.75 hybrid)
- 5-year hardware refresh
- No major disruptions

Out of Scope

- Customer-facing systems (ATMs, public Wi-Fi)
- Full data center integration
- Ongoing software dev
- Inflation beyond 3%/yr or forex fluctuations
- Cost of property
- Cost of Electricty

COST FRAMEWORK

To support consistent decision-making, this analysis is built on a simple cost framework that distinguishes between:

- Capital expenses (CapEx)
- Operating expenses (OpEx)
- Total cost of ownership (TCO)

The framework also accounts for both direct and indirect costs, so that the numbers reflect the full impact of opening a new branch IT environment.

Capital Expenses

Capital expenses are the one-time or infrequent costs associated with purchasing and installing hardware, software licenses, and other long-lived assets required to set up the branch IT environment. For both the traditional and hybrid models, CapEx includes:

- Servers, storage, networking gear, and endpoints (PCs, thin clients, printers, etc.).
- Initial cloud setup costs (landing-zone configuration, connectivity, and migration tooling).
- Implementation and project labour related to the initial deployment.

These costs are typically incurred in Year 0 or early in the project, and for the purpose of this analysis they are treated as a single upfront investment, even though some items may be purchased over a short period

Operating Expenses

Operating expenses are the recurring costs of running and maintaining the IT environment over time. OpEx includes:

- Cloud subscription fees (compute, storage, networking, and managed services).
- Power, cooling, and space-related facility costs for on-prem hardware.
- Software licenses and support contracts (e.g., OS, security, backup, and monitoring).
- Ongoing operations and maintenance labour (monitoring, patching, troubleshooting, and minor upgrades).

For the traditional model, OpEx is relatively stable but still includes hardware support and maintenance. For the hybrid model, more OpEx is shifted into cloud subscriptions, which can scale up or down with business needs.

Total Cost Ownership

Total Cost of Ownership combines CapEx and OpEx over the full lifecycle of the infrastructure, providing a clearer picture of long-term cost than looking at up-front or annual costs in isolation. In this report:

- 5-year TCO captures the full cost of deploying and running the new branch IT environment for half a decade, including hardware refresh cycles and major support renewals where applicable.
- 10-year TCO extends the view to ten years, highlighting the impact of depreciation, end-of-life replacements, and long-term operational commitments.

By comparing TCO across the traditional and hybrid models, the bank can assess not only which option is cheaper initially, but also which better supports long-term flexibility, scalability, and control within a branch-expansion strategy.

TRADITIONAL VS HYBRID

Traditional Infrastructure

For the new branch, the traditional model assumes:

- One or more on-prem servers for core banking applications, file services, and identity services.
- Local storage and backup appliances.
- Physical switches, routers, and Wi-Fi infrastructure.
- Desktop PCs or thin clients for the 20 internal users.

All hardware is owned and operated by the bank, installed in a branch-level server room or equipment closet, with connectivity back to the bank's core data centre.

Cost structure

The analysis breaks the traditional model into the following cost categories:

Hardware CapEx (Year 1):
Servers, storage, networking gear, backup appliances, UPS, cabling, and desktop endpoints for the 50-user environment. This reflects the one-time purchase cost plus any installation services required to set up the branch.

Labour and implementation cost:
Estimated hours for network cabling, server and switch configuration, endpoint deployment, and integration with the bank's existing IT environment. These are treated as part of the initial CapEx but can be shown as a separate line item for clarity.

Annual OpEx (ongoing):

- Power, cooling, and basic space-related costs for the branch server environment.
- Software licenses and support contracts (OS, antivirus, backup software, monitoring).
- Maintenance and patching labour, plus minor hardware repairs or replacements.

By accumulating CapEx and yearly OpEx over 5 and 10 years, the traditional model reveals how recurring support and periodic refresh cycles (for example, replacing servers every 5 years) influence the total cost of ownership for a branch-based on-prem setup

Hybrid Cloud Infrastructure

For the same 20-user branch, the hybrid model assumes:
·A smaller on-prem footprint (for example, one consolidated server or appliance for branch-specific apps and local storage).
·Core banking and shared services hosted in the cloud (Azure, or another public cloud provider).
·Cloud-based backup, identity, and collaboration services.
·Branch-level networking and Wi-Fi, similar to the traditional model, but with lighter local compute and storage requirements.
Workloads that are sensitive or latency-critical (e.g., local transaction processing or regulatory reporting) remain on-prem, while less critical or burst-prone workloads leverage the cloud's elasticity.

Cost structure

The hybrid model splits costs between on-prem components and cloud components:

On-prem hardware CapEx (Year 1):
Reduced hardware set (consolidated server, local storage, networking gear, UPS, and endpoints for 50 users). This is typically lower than the traditional model because the cloud carries more of the workload.

Cloud setup and migration CapEx:
One-time configuration costs for connectivity (ExpressRoute/VPN), landing-zone setup, and migration tools. This is shown as a small CapEx uplift to enable the hybrid model.

Annual OpEx (ongoing):

- Cloud subscription fees (compute, storage, networking, managed services, and SaaS tools).
- On-prem power, cooling, and support for the reduced local infrastructure.
- Software licenses and cloud-based management tools.
- Ongoing monitoring, patching, and minor cloud-cost-optimization effort.

Capital Expenditure: Hardware Cost
Traditional Infrastructure

QTY.	Device	Cost	Total
4	Cisco Integrated Services Router 1131	$2736.52	$12,259.61
4	Cisco Catalyst 1200-8P-E-2G Switch	$409.28	$1833.57
8	Cisco Business 220 Series CBS220-24T-4G Switch - 24 ports	$328.99	$2947.75
20	Lenovo ThinkStation P3 Gen 2 - tower Core Ultra 5	$2929.00	$65,609.60
20	Acer Vero V247Y G 24" Class LCD Monitor	$164.99	$3695.78
20	Lenovo Essential - keyboard - QWERTY	$29.99	$671.78
20	Lenovo Basic - mouse - USB	$24.43	$547.23
20	Cisco IP Phone 7841	$719.26	$16,111.42
7	Ubiquiti DreamWall - security appliance - Wi-Fi 6	$1481.99	$11,618.80
14	HP LaserJet Enterprise M507 M507n Desktop Laser Printer	$1098.88	$17,232.16
1	QNAP Thunderbolt 4 8-Bay Network Attached Storage App.	$7151.99	$8010.22
8	Seagate SkyHawk AI ST24000VE002 - hard drive - 24 TB	$727.99	$6522.79
1	1000' spool of cable - Eaton Tripp Lite Series Cat8 25G/40G	$2592.32	$2903.39
		Total	$149,964.10

Capital Expenditure: Hardware Cost
Hybrid Infrastructure

QTY.	Device	Cost	Total
1	Supermicro server - Dual xeon 6505p	$19,112.26	$23,905.75
20	Lenovo ThinkStation P3 Gen 2 - tower Core Ultra 5	$2929.00	$65,609.60
20	Acer Vero V247Y G 24" Class LCD Monitor	$164.99	$3695.78
20	Lenovo Essential - keyboard - QWERTY	$29.99	$671.78
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1	1000' spool of cable - Eaton Tripp Lite Series Cat8 25G/40G	$2592.32	$2903.39
		Total	$100,895.88

Capital Expenditure: Labor Cost (Initial Setup)
Traditional Infrastructure

Role	Hours Quoted	Low Total CAD	High Total CAD
Cabling Technician	300	$21,000	$26,400
Network Engineer	240	$25,200	$29,040
System Administrator	120	$11,040	$12,600
Project Manager	80	$9120	$10,610
Total	740	$66,360	$78,200

Capital Expenditure: Labor Cost (Initial Setup)
Hybrid Infrastructure

Role	Hours Quoted	Low Total CAD	High Total CAD
Cabling Technician	150	$10,500	$13,200
System Administrator	80	$7360	$19,300
Network Engineer	160	$16,800	$8400
Cloud Architect	100	$12,000	$14,000
Project Manager	60	$6840	$7620
Total	550	$53,500	$62,580

Operational Expenditure: Labor Cost
Traditional Infrastructure

Role	FTE	Low Total CAD	High Total CAD
System Administrator	0.75	$140,000	$170,000
Network Engineer	1.0	$83,000	$105,000
Cloud Network Architect	0.0	$0	$0
MSP (Outsourced)	0.25	$15,000	$23,000
Total	2.0	$238,000	$298,000

Operational Expenditure: Labor Cost
Hybrid Infrastructure

Role	FTE	Low Total CAD	High Total CAD
System Administrator	0.50	$55,000	$70,000
Network Engineer	0.75	$105,000	$128,000
Cloud Network Architect	0.25	$40,000	$48,000
MSP (Outsourced)	0.25	$15,000	$23,000
Total	1.75	$215,000	$269,000

FTE - Full Time Equivalent
The measure in allocated equivalent for maintaining branch's IT infrastructure

NON-FINANCIAL FACTORS

While financial metrics provide the core comparison, this section evaluates the qualitative advantages, risks, and intangibles of each model for a new bank branch. These factors include operational efficiency, compliance, scalability, and strategic flexibility.

Traditional Infrastructure

Direct Benefits:

- Complete control over hardware and data location, simplifying compliance audits and regulatory reporting.
- Predictable performance for local branch operations with minimal latency for critical transaction processing.

Indirect & Intangible Benefits:

- Familiar operational model for existing IT staff, reducing training requirements.
- Perceived security advantage through physical isolation from external networks.

Risks & Opportunity Costs:

- Limited scalability for rapid branch growth or new service deployment without additional hardware purchases.
- Higher downtime risk from single-point failures (no built-in cloud redundancy).
- Missed opportunity for innovation like AI analytics or rapid feature updates.

Hybrid Cloud Infrastructure

Direct Benefits:

- Elastic scalability to handle peak transaction volumes or seasonal demand without over-provisioning hardware.
- Faster deployment of new services (e.g., mobile banking features) through cloud-native tools.

Indirect & Intangible Benefits:

- Reduced IT staffing burden as cloud providers handle infrastructure maintenance and patching.
- Access to modern cloud services (AI analytics, real-time fraud detection) without custom development.
- Enhanced business agility for responding to market changes or regulatory updates.

Risks & Opportunity Costs:

- Shared responsibility model requires clear cloud security governance and vendor management.
- Potential vendor lock-in and data egress costs if migrating between cloud providers.
- Complexity of managing dual environments (on-prem + cloud) requires hybrid skills.

RECOMMENDATION

Recommendation for Traditional On-Premise Infrastructure
Choose the Traditional On-Premise model if the bank's primary priority is maximum local control and a familiar operating model, and if cloud adoption is constrained by policy or regulation.

- This model is appropriate when regulators or internal risk teams require that most systems and data remain physically on-site, or where branch connectivity to the cloud is unreliable.
- It fits organizations with an existing strong on-prem operations team, standardized hardware procurement processes, and long depreciation cycles.
- However, the bank should accept a higher total cost of ownership over 5 and 10 years, slower deployment of new services, and limited scalability for future branch expansion compared to hybrid.

If the bank selects this option, the recommendation is to:
- Standardize a branch "hardware kit" (routers, switches, PCs, NAS, UPS) for all new branches to simplify procurement and support.
- Invest in robust local redundancy (RAID, dual PSUs, backup power) to mitigate downtime from single-site failures.

8.2 Recommendation for Hybrid Cloud Infrastructure
Choose the Hybrid Cloud model as the preferred option for new branch rollouts, especially when the bank aims to optimize long-term cost, scalability, and innovation.

- Hybrid reduces initial CapEx and lowers both 5-year and 10-year TCO, making it financially more attractive for branch expansion.
- It supports elastic scaling for transaction peaks, faster rollout of new digital services, and easier integration with modern cloud capabilities
- Sensitive workloads and regulated data can remain on-prem at the branch or central data centre, while less critical services move to the cloud under a clear shared-responsibility model.

If the bank selects this option, the recommendation is to:
- Establish a standard hybrid reference architecture (on-prem branch node + cloud landing zone) and reuse it for future branches.

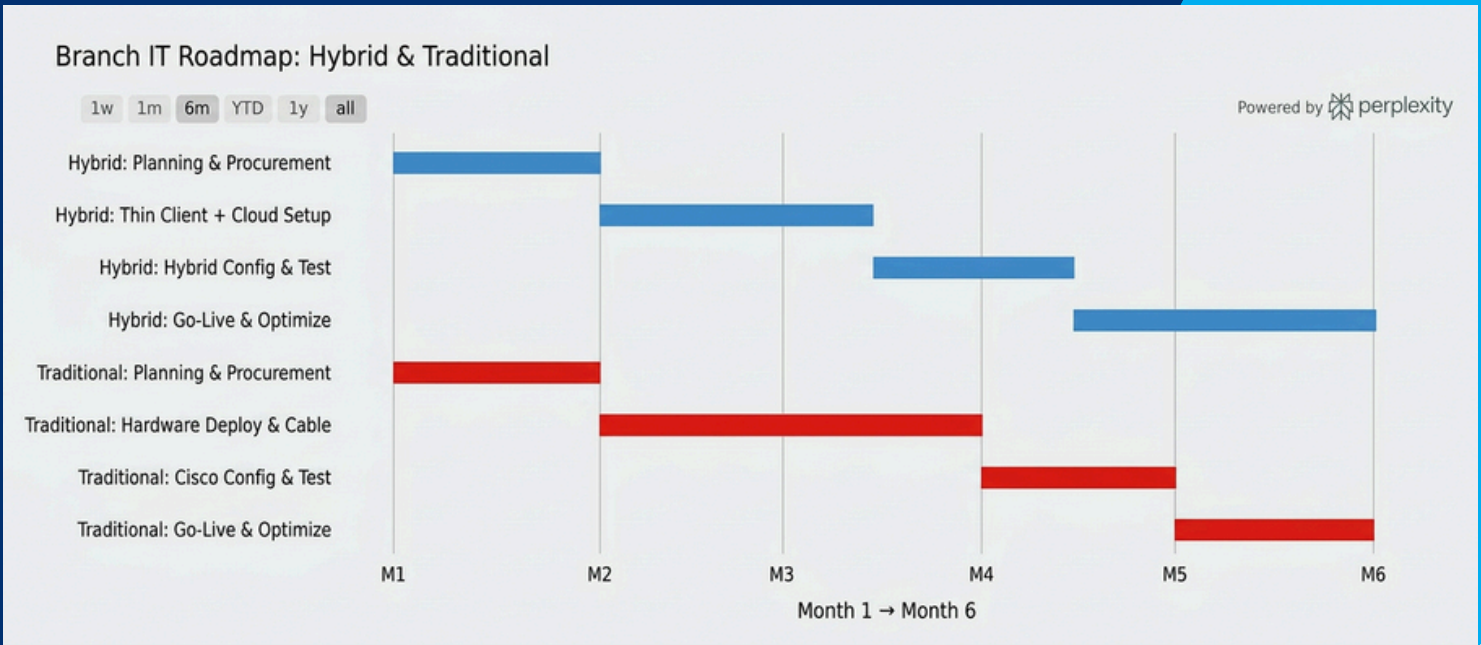
ROADMAP (BOTH MODELS)

Phase 1: Planning & Procurement (Month 1)
Traditional: Procure full hardware kit (servers, PCs, Cisco gear)
Hybrid: Procure lightweight hardware (Supermicro, thin clients) + cloud subscription

Phase 2: Hardware & Network Deployment (Months 1–2.5)
Traditional: Server rack install, extensive cabling, Cisco router/switch config (2 months)
Hybrid: Simplified cabling, thin client setup, cloud connectivity (1.5 months)

Phase 3: Configuration & Testing (Months 2–3.5)
Traditional: Full Cisco network config, server OS, endpoint imaging (1.5 months)
Hybrid: Hybrid config (on-prem + cloud), endpoint testing (1 month)

Phase 4: Go-Live & Optimization (Months 3–5)
Both: Staff training, final testing, branch opening, 30-day monitoring and tuning



Total Timeline: ~4.5–5 months for both models. Hybrid is slightly faster due to less hardware complexity, but both deliver a fully operational branch within the same business quarter.